Data Structures and Algorithms CA1

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Course: BSc (Honours) in Computing - evening

Year: 2

Project content:

This is an individual project. The aims of this project are to design and implement a Java application using abstract data types that were taught during the classes. The application must include interfaces, a GUI, 3 abstract data types, inheritance, polymorphism, Object data, and CRUD operations. Along with the project, there must be a written report included containing the theme of the application based on the Smart Docklands Pilot Call webpage. The report must describe the application and the problem it solves, include a class diagram, be committed to a GitHub repo, and include the link to the GitHub repo.

Academic Integrity Form:

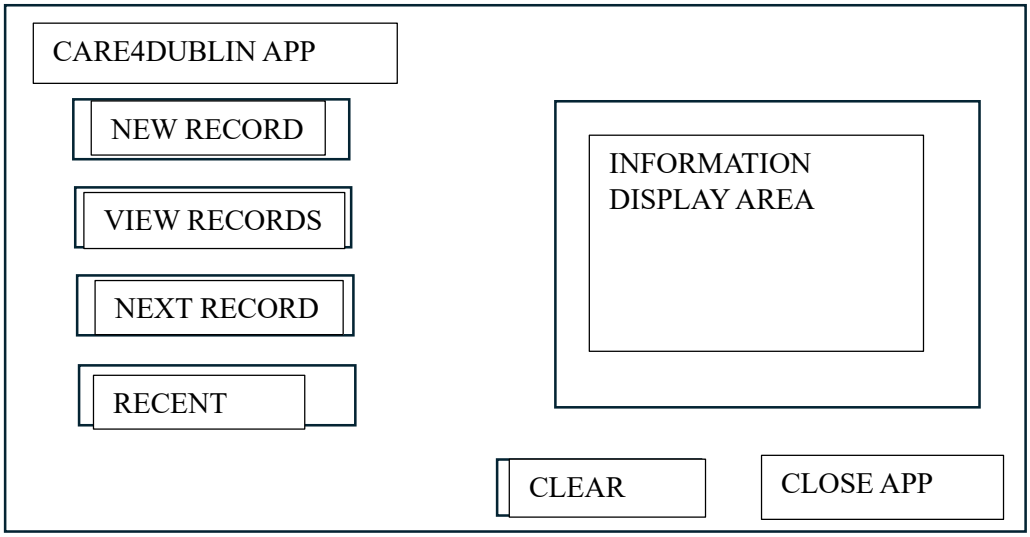
Theme chosen:

Urban Greening/Biodiversity is my chosen theme for this project.

Description of Application:

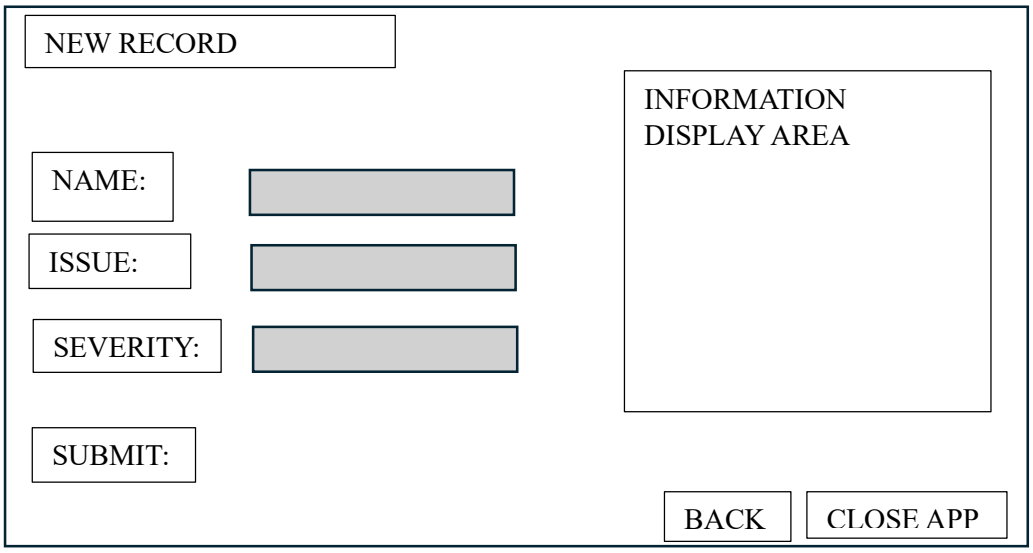
The application name is Care4Dublin. The application is meant to be used by the Dublin City Council to store reports sent in by locals about environmental issues. The application has 2 pages for simplicity and user friendliness. Page 1 is the landing page: consisting of the title, 4 interactive buttons, an information area, a clear info area button, and a close app button. The New Record button brings the user to the 2nd page. The view records button displays the records in the information area. The next record button will display the next record stored based on the highest priority. The recently resolved button shows the latest issue that has been resolved. The clear button clears the display area. Lastly, the close app button closes the application.

(Planned Landing Page)



The new record page will allow a member of the Dublin City Council to input a new record of an environmental issue within the docklands that was reported by a local. The page will consist of a title, 3 input fields – name, issue, and severity, a submit button, an information display area, a back button, and a close app button. This page aims to be simple to use for any level of computational user. The submit button accepts input from the user and stores it in the Abstract Data Types and then displays a message in the display area. The back button brings the user back to the landing page. Lastly, the close app button closes the application.

(Planned New Record Page)



Problem Solved in Relation to the Theme:

This application aims to better the communication between the locals and the City Council in relation to the Environment. Locals contact the Council via E-Mail or Phone Number addressing issues in their area (in this case environmental issues). Then an administrator working alongside the environmental department can upload the individual's name, issue addressed, and the severity of the issue thus allowing the Council to take swifter action to resolve the issue.

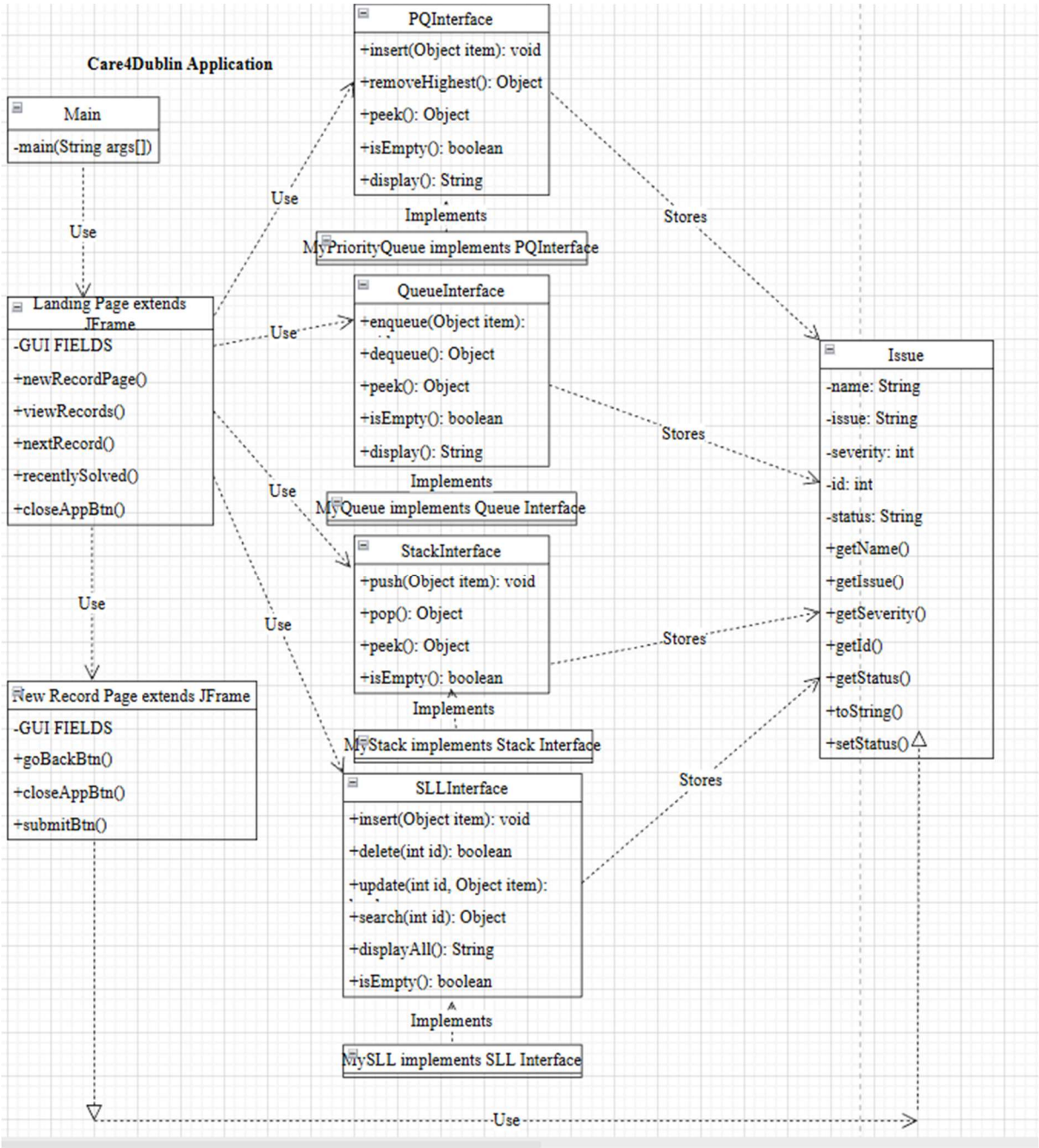
How the Application fits with the Theme:

The correlation between this Application and the theme of Urban Greening/Biodiversity lies within the applications usage to accurately depict the unresolved issues addressed by those who seek correction within their living space. At the heart of this project is the care I have for the environment, seeking a more efficient way at tackling the daily environmental issues that are present/occur in urban areas. The project is built to help prioritise people, animals, water, the air, and nature.

Abstract Data Types Used:

1. Singly Linked List (Mandatory): Will store the issues.
2. Priority Queue: New Record page will incorporate Priority Queue with the severity input (Highest Priority First).
3. Queue: Reports pending will be created using a Queue (FIFO).
4. Stack: Recently resolved will be accessed via the stack principle (LIFO).

Class Diagram (Draw.io):



Public GitHub Repository Link:

https://github.com/Aidas04/Smart_Docklands_Project_Aidas.git